

NAME: \_\_\_\_\_

PERIOD: \_\_\_\_\_

## Which Error Can the District Afford?

AP Statistics · Unit 3 · Topic 3.8

### [STAR] THE PROBLEM

A district checks an emergency-alert system. Let  $p$  be the proportion of alerts that arrive late. It tests  $H_0 : p = 0.05$  versus  $H_a : p > 0.05$ , replacing the system if it rejects  $H_0$ . Both plans use  $n = 200$ . The Safety office wants to avoid keeping a system with a 10% late rate. The Budget office wants to avoid replacing a system with a 5% late rate. The offices have not agreed how to compare these costs.

**Dev:** “Use A. Smaller  $\alpha$  lowers both error risks.”

**Lin:** “Use B. Its higher power has no cost when safety matters.”

Test plans ( $n = 200$ )

| Plan | $\alpha$ | Power at $p = 0.10$ |
|------|----------|---------------------|
| A    | 0.010    | 0.748               |
| B    | 0.050    | 0.877               |

### FIRST TAKE — one sentence before you work the parts

Which plan would you try first? Name the office you prioritized and one competing cost.

### [BOOK] ERROR PROBABILITIES TRADE OFF (keep on the page)

A Type I error rejects a true  $H_0$ , and its probability is  $\alpha$ . A Type II error fails to reject a false  $H_0$ ;  $\beta = 1 - \text{power}$ . With sample size fixed, increasing  $\alpha$  generally raises power and lowers  $\beta$ , but it also raises the false-positive risk. Increasing sample size can raise power while keeping the chosen  $\alpha$  fixed. Choose a plan by naming the real consequence of each error.

**Work (a)–(c).**

**DOK 1** (a) Identify what the table's power at  $p = 0.10$  means in this setting.

**DOK 2** (b) Calculate the Type II error probability  $\beta$  for each plan. State each plan's Type I error probability.

**DOK 3** (c) In 4–5 sentences, recommend a plan for one office using both error risks. Explain the consequence of each error and what Dev and Lin each overlook. State why the other office might choose differently.

**Frame:** For the \_\_\_\_\_ office, I recommend \_\_\_\_\_ because \_\_\_\_\_.

**Frame:** That plan risks \_\_\_\_\_ when \_\_\_\_\_; the other office may prioritize \_\_\_\_\_.

**Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.**